Link Layer & LANs Definitions (Chapter 6)

6.1 Introduction to the Link Layer

Node Any device in a network, including hosts and routers.
Link A communication channel connecting two adjacent nodes.
Frame The Layer-2 (link layer) packet. It encapsulates a network-layer datagram.
Link Layer Responsibility To transfer a datagram from one node to a physically adjacent node over a single link (a hop-by-hop service).
Framing The link-layer service of encapsulating a datagram into a frame by adding a header and trailer.
Link Access The service of coordinating transmissions on a shared medium. Managed by *Multiple Access Protocols*.
Hop-by-Hop Reliable Delivery A link-layer service that ensures a frame is successfully transmitted across a single link. Crucial for high-error links like wireless.
NIC (Network Interface Card) The hardware (and associated software/firmware) in a host or router that implements the link-layer and physical-layer protocols.

6.2 Error-Detection and Correction Techniques

EDC (Error Detection & Correction Bits) Redundant bits added to a frame to detect or correct errors.
Parity Checking A simple error-detection scheme.
- **Single Bit Parity:** Adds one bit to a block of data to make the total number of 1s either even or odd. Can detect single-bit errors.
- **Two-Dimensional Parity:** Arranges data in a grid and computes parity for each row and column. Can detect and correct single-bit errors.

CRC (Cyclic Redundancy Check) A powerful error-detection technique used in Ethernet and Wi-Fi. It treats data as a polynomial and uses a pre-agreed **Generator (G)** to compute a remainder, which is appended to the frame. The receiver checks if the entire frame is divisible by G.

6.3 Multiple Access Links and Protocols

Shared Medium A communication channel where multiple nodes are connected and can transmit, creating the possibility of collisions.
Collision Occurs when two or more nodes transmit on a shared medium at the same time, resulting in a garbled, unusable signal.
Multiple Access Protocol A distributed algorithm that coordinates how nodes share a common channel.
Channel Partitioning Protocols Protocols that divide the channel's resources into dedicated, collision-free pieces.
- **TDMA (Time Division Multiple Access):** Allocates a fixed time slot to each node.
- **FDMA (Frequency Division Multiple Access):** Allocates a fixed frequency band to each node.

Random Access Protocols Protocols where nodes transmit at will and the protocol defines how to detect and recover from collisions.
- **ALOHA / Slotted ALOHA:** Simple protocols where nodes transmit immediately or at the start of the next slot. Low efficiency due to high collision rates.
- **CSMA (Carrier Sense Multiple Access):** "Listen before you talk." A node first senses the channel to see if it is idle before transmitting.
- **CSMA/CD (Collision Detection):** "Listen while you talk." Used in classic Ethernet. A node senses the channel and also monitors its own transmission. If a collision is detected, it aborts transmission and enters an **exponential backoff** period.

Taking Turns Protocols Protocols that create a coordinated, collision-free sharing of the channel (e.g., **Polling**, **Token Passing**).

6.4 Switched Local Area Networks (LANs)

MAC Address A 48-bit, unique, permanent "physical" address burned into a NIC. Used for addressing frames to other nodes on the *same local network*. It is a "flat" address.
ARP (Address Resolution Protocol) The protocol that translates a network-layer IP address into a link-layer MAC address.
- **ARP Table:** A cache in each host/router that stores recent IP-to-MAC address mappings.
- **ARP Query:** A broadcast message sent to all nodes on a LAN, asking "Who has this IP address?".
- **ARP Response:** A unicast message sent directly back to the querier, providing the requested MAC address.

Ethernet The dominant wired LAN technology. Modern Ethernet uses a star topology with a central switch.
- **Ethernet Frame:** Contains a Preamble, Dest/Src MAC Addresses, Type field, Payload (IP datagram), and a 32-bit CRC.

- **Unreliable, Connectionless:** Ethernet provides no acknowledgements or handshaking.

Link-Layer Switch A Layer-2 device that receives and forwards Ethernet frames. It is a "store-and-forward" device.

- **Transparent:** Hosts on the LAN are unaware of the switch's presence.
- **Self-Learning:** The switch automatically builds its forwarding table by inspecting the **source MAC address** of incoming frames.
- **Switch Table:** Maps destination MAC addresses to the switch's output interfaces (ports).
- **Forwarding/Filtering:** If a destination MAC is in the table, the switch forwards the frame only to the correct port (selective forwarding).
- **Flooding:** If a destination MAC is *not* in the table, the switch broadcasts the frame to all ports except the one it arrived on.

6.5 Link Virtualization

MPLS (Multiprotocol Label Switching) A link-layer technology used in ISP backbones that forwards packets based on short, fixed-length **labels** instead of IP addresses. This allows for **traffic engineering** by creating pre-determined forwarding paths called **Label-Switched Paths (LSPs)**.